DSA Project Report

Expense Management System Using Arrays

Project Description

This project is a console-based Expense Management System developed in C++. The system allows users to add multiple expenses under predefined categories such as Food, Transport, Study, and Miscellaneous. All data is stored using arrays, and the system provides basic searching and total calculation features.

Features of the System

- Added multiple expense categories
- Predefined expense categories
- Automatic total expense calculation
- Search expenses by category
- Search expenses by amount

Program Flow

- User selects an option from the menu
- Expenses are added under a selected category
- Data is stored in arrays
- Searching is performed when required
- Total expenses are displayed
- Program exits on user choice

Data Structures Used

Array

Arrays are used to store expense amounts and category information in an organized manner.

Algorithms Used

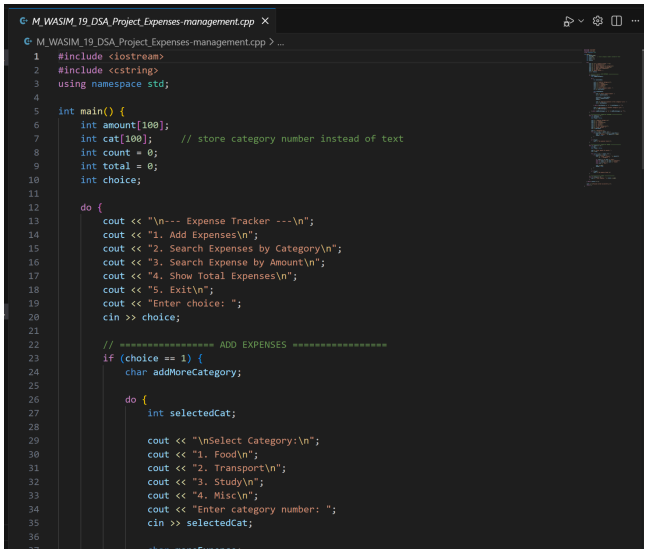
Linear Search

Linear search is used to find expenses by category or amount.

Conclusion

This mini project demonstrates the use of basic Data Structures and Algorithms concepts using C++. It provides a simple solution for expense management and helps in understanding arrays and searching techniques.

Screenshots



```cpp
#include <iostream>
#include <string>
using namespace std;

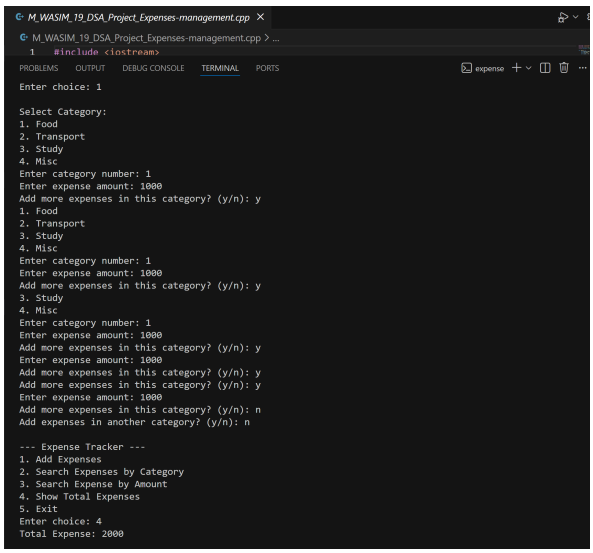
int main() {
    int amount[100];
    int cat[100];     // store category number instead of text
    int count = 0;
    int total = 0;
    int choice;

    do {
        cout << "\n--- Expense Tracker ---\n";
        cout << "1. Add Expenses\n";
        cout << "2. Search Expenses by Category\n";
        cout << "3. Search Expense by Amount\n";
        cout << "4. Show Total Expenses\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        // ================ ADD EXPENSES ================
        if (choice == 1) {
            char addMoreCategory;

            do {
                int selectedCat;

                cout << "\nSelect Category:\n";
                cout << "1. Food\n";
                cout << "2. Transport\n";
                cout << "3. Study\n";
                cout << "4. Misc\n";
                cout << "Enter category number: ";
                cin >> selectedCat;
```



```
Enter choice: 1

Select Category:
1. Food
2. Transport
3. Study
4. Misc
Enter category number: 1
Enter expense amount: 1000
Add more expenses in this category? (y/n): y
1. Food
2. Transport
3. Study
4. Misc
Enter category number: 1
Enter expense amount: 1000
Add more expenses in this category? (y/n): y
3. Study
4. Misc
Enter category number: 1
Enter expense amount: 1000
Add more expenses in this category? (y/n): y
Enter expense amount: 1000
Add more expenses in this category? (y/n): y
Add more expenses in this category? (y/n): y
Enter expense amount: 1000
Add more expenses in this category? (y/n): n
Add expenses in another category? (y/n): n

--- Expense Tracker ---
1. Add Expenses
2. Search Expenses by Category
3. Search Expense by Amount
4. Show Total Expenses
5. Exit
Enter choice: 4
Total Expense: 2000
```


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